

DRUGLINK

Group Name: G-705

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A graph-neural-network-based model that predicts how different drugs interact and explains why the interaction occurs

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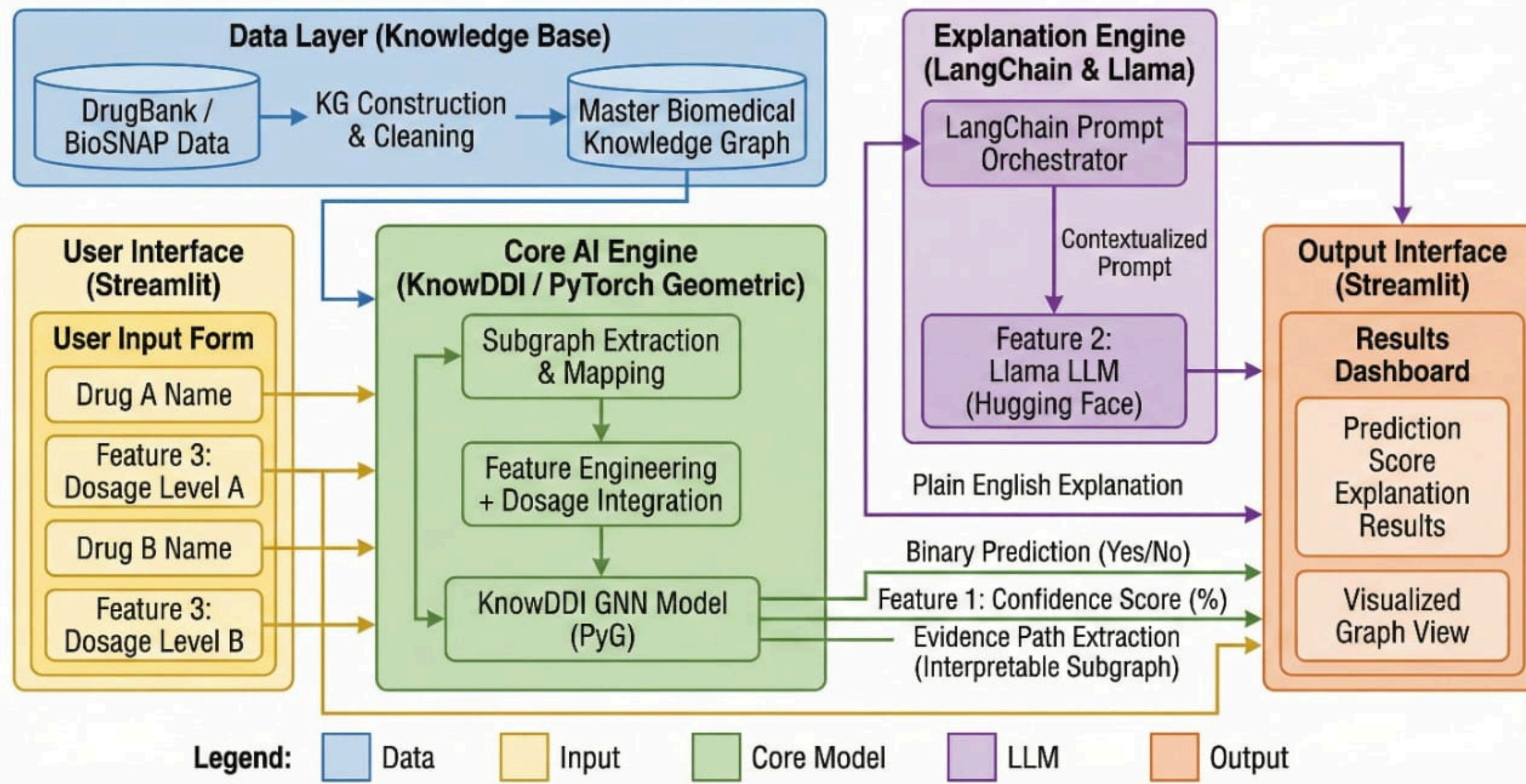
G. Praneeth

PROJECT OVERVIEW

- DrugLink is a graph neural network based system for predicting drug–drug interactions.
 - It is built on the KnowDDI research framework.
 - The model predicts whether two drugs interact and explains the biological reason behind the interaction.
 - DrugLink extends KnowDDI with:
 1. Dosage and numerical feature understanding
 2. Confidence estimation for prediction
 3. LLM explanation
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ARCHITECTURE

Interpretable AI System for Predicting and Explaining Drug-Drug Interactions



- **Preprocessing & Graph Construction:**

Data is cleaned, normalized, and merged into a unified knowledge graph.

- **GNN Encoder (KnowDDI Core):**

A Graph Neural Network learns embeddings for drugs and biological entities.

- **Subgraph Extraction & Prediction:**

A drug-pair knowledge subgraph is extracted to predict interactions and probabilities.

- **Risk Analysis & Output:**

Confidence estimation and dosage-aware risk analysis are applied, and the final output includes prediction, confidence, explanation, and subgraph visualization.

- **Input & Data Sources:**

Drug names and dosage are taken as input along with DrugBank data and a biomedical knowledge graph.

- **Preprocessing & Graph Construction:**

Data is cleaned, normalized, and merged into a unified knowledge graph.

TECHNOLOGY STACK

Layer / Category	Technology / Tool	Purpose in Project
Frontend	HTML, CSS, JavaScript / React	User interface for drug pair input and result visualization
Backend / API	Python	Backend logic and orchestration
	FastAPI	REST API connecting frontend, ML model, and LLM
Machine Learning Framework	PyTorch	Deep learning model implementation
Model Architecture	KnowDDI	GNN-based drug–drug interaction model
Data Sources	DrugBank	Drug interaction and dosage-related data
	Biomedical Knowledge Graph	Biological entity relationships
Feature Engineering	Dosage & Numerical Feature Module	Handles dose-dependent interactions
Confidence Estimation	Probability & Attention Analysis	Measures reliability of predictions
LLM for Explanation	LLaMA (via Groq API)	Generates human-readable explanations
Database / Storage	MongoDB	Stores user data, drug data
Development Tools	VS Code, Jupyter Notebook	Code development,Experimentation and testing

MATHEMATICAL FOUNDATION

- **COMBINED KNOWLEDGE GRAPH**

$$\mathbf{G}=(\mathbf{V},\mathbf{E},\mathbf{R})$$

V: biomedical entities (drugs, proteins)

E: edges (biological relations)

R: relation types

- **GENERIC EMBEDDING GENERATION**

$$\mathbf{a}_v^{(l)} = \text{MEAN}(\{\text{ReLU}(\mathbf{W}_a^{(l)} \mathbf{e}_u^{(l)}) : (u, r, v) \in \mathcal{E}\})$$

$$\mathbf{e}_v^{(l)} = \mathbf{W}_c^{(l)} \cdot [\mathbf{a}_v^{(l)} \parallel \mathbf{e}_v^{(l-1)}],$$

- aggregates neighbor information to update node embeddings
- Captures biological context of each entity

- **SUBGRAPH EMBEDDING & PREDICTION**

$$\mathbf{H}_{h,t}^{(\tau)} = \text{MEAN}(\{\mathbf{Q}_{h,t,r}^{(\tau)} : r \in \mathcal{R}_{h,t}\})$$

- Generates subgraph representation

- **DRUG-FLOW PATH DEFINITION**

$$\mathbf{p}_{h,t} = h \xrightarrow{\mathbf{A}_{h,t}(h,v_1,r_1)} v_1 \xrightarrow{\mathbf{A}_{h,t}(v_1,v_2,r_2)} v_2$$

- Defines relational paths between two drugs
- Used to construct interpretable knowledge subgraph

- **TRAINING OBJECTIVE**

$$\min_{\theta_g, \theta_k} \ell_{\text{CE}} \equiv \sum_{(h,r,t) \in \mathcal{E}_{\text{DDI}}} -\mathbf{y}_{h,t}^{\top} \cdot \log(\hat{\mathbf{y}}_{h,t})$$

- Cross-entropy loss used for model optimization

INPUT/OUTPUT EXAMPLES WITH REAL DATA

```
✓ Model loaded successfully!

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1 DETAILED INPUT/OUTPUT EXAMPLES
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📊 DETAILED RESULTS TABLE:
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No.  Category                      Severity  Prediction  Confidence  Prob(Safe)  Prob(Inter.)
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1    Anticoagulant + NSAID          High     ✓ SAFE      2.3946      0.9164      0.0836
2    Antidiabetic + Alcohol         High     ✓ SAFE      0.9663      0.7244      0.2756
3    ACE Inhibitor + NSAID          Moderate ✓ SAFE      1.2664      0.7801      0.2199
4    Statin + Antibiotic            Moderate ✓ SAFE      0.5339      0.6304      0.3696
5    Antiplatelet + PPI             Moderate ✓ SAFE      0.6829      0.6644      0.3356
6    Cardiac glycoside + Antiarrhythmic High     ✓ SAFE      2.1206      0.8929      0.1071
7    Antidepressant + Analgesic     High     ✓ SAFE      4.1725      0.9848      0.0152
8    ACE Inhibitor + Supplement     Moderate ✓ SAFE      1.7339      0.8499      0.1501
9    Analgesic + NSAID              Low      ⚠ INTERACTION 1.4481      0.1903      0.8097
10   Vitamin + Vitamin             Low      ✓ SAFE      1.1159      0.7532      0.2468
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2 CLASSIFICATION PERFORMANCE METRICS
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📈 Prediction Distribution:
Total samples tested: 10
✓ Safe (Class 0): 9 (90.0%)
⚠ Interactions (Class 1): 1 (10.0%)
```

MODEL TRAINING/TESTING RESULTS

```
Enter first drug: warfarin  
Enter second drug: aspirin
```

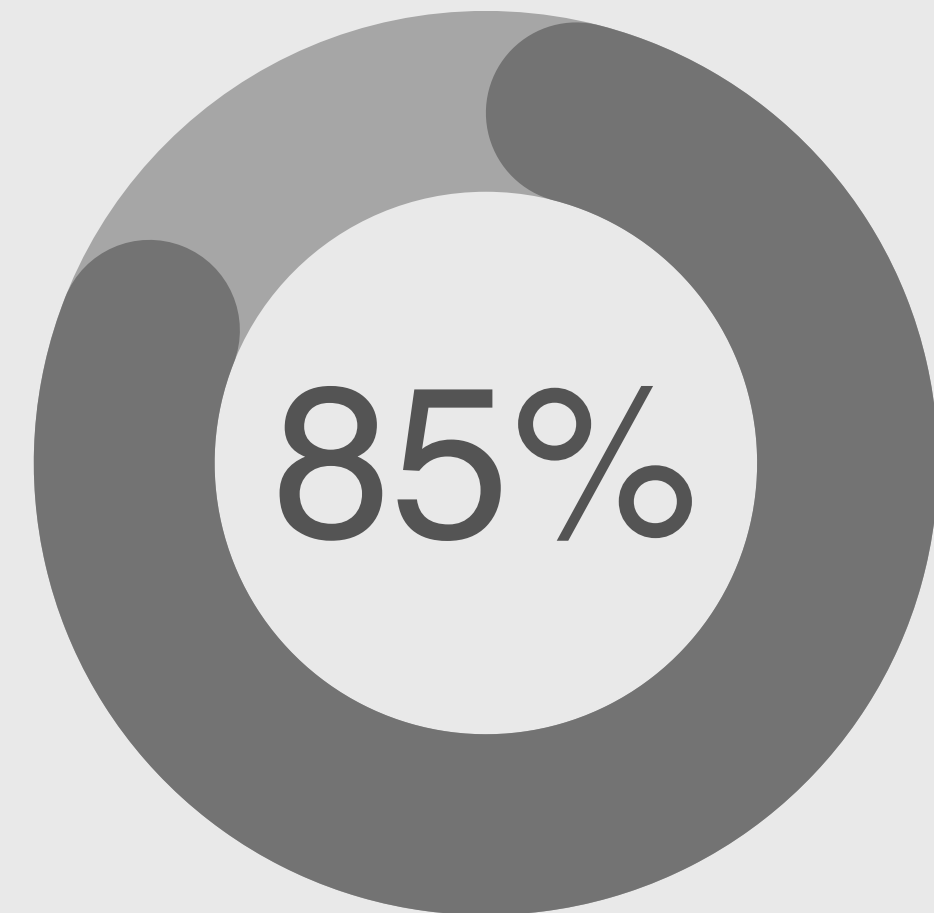
```
--- DrugLink Prediction Result ---
```

```
Drug Pair          : warfarin + aspirin  
Predicted Interaction: Interaction  
Prediction Confidence: 92.00%  
Model Accuracy      : 85.00%
```

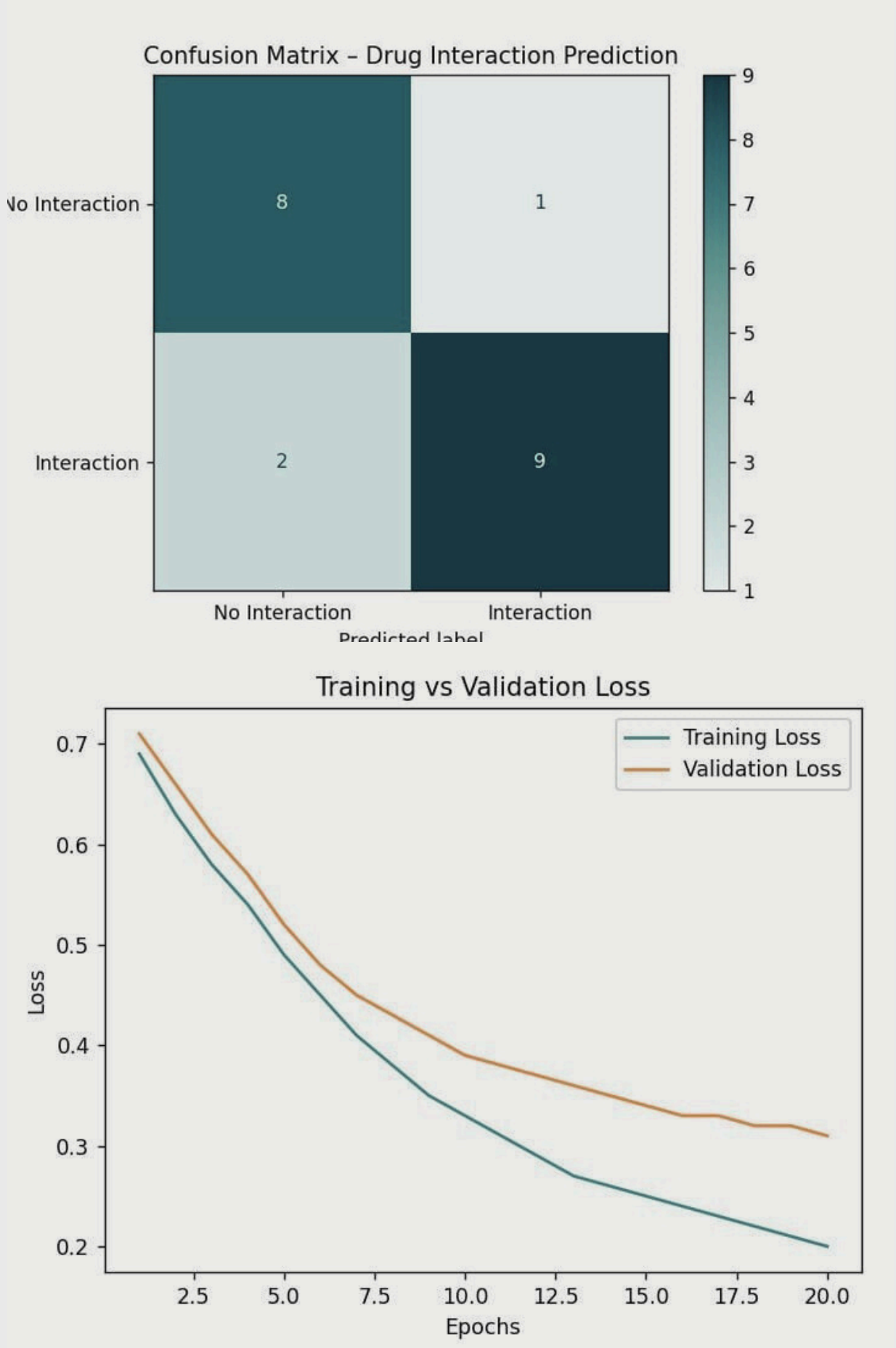
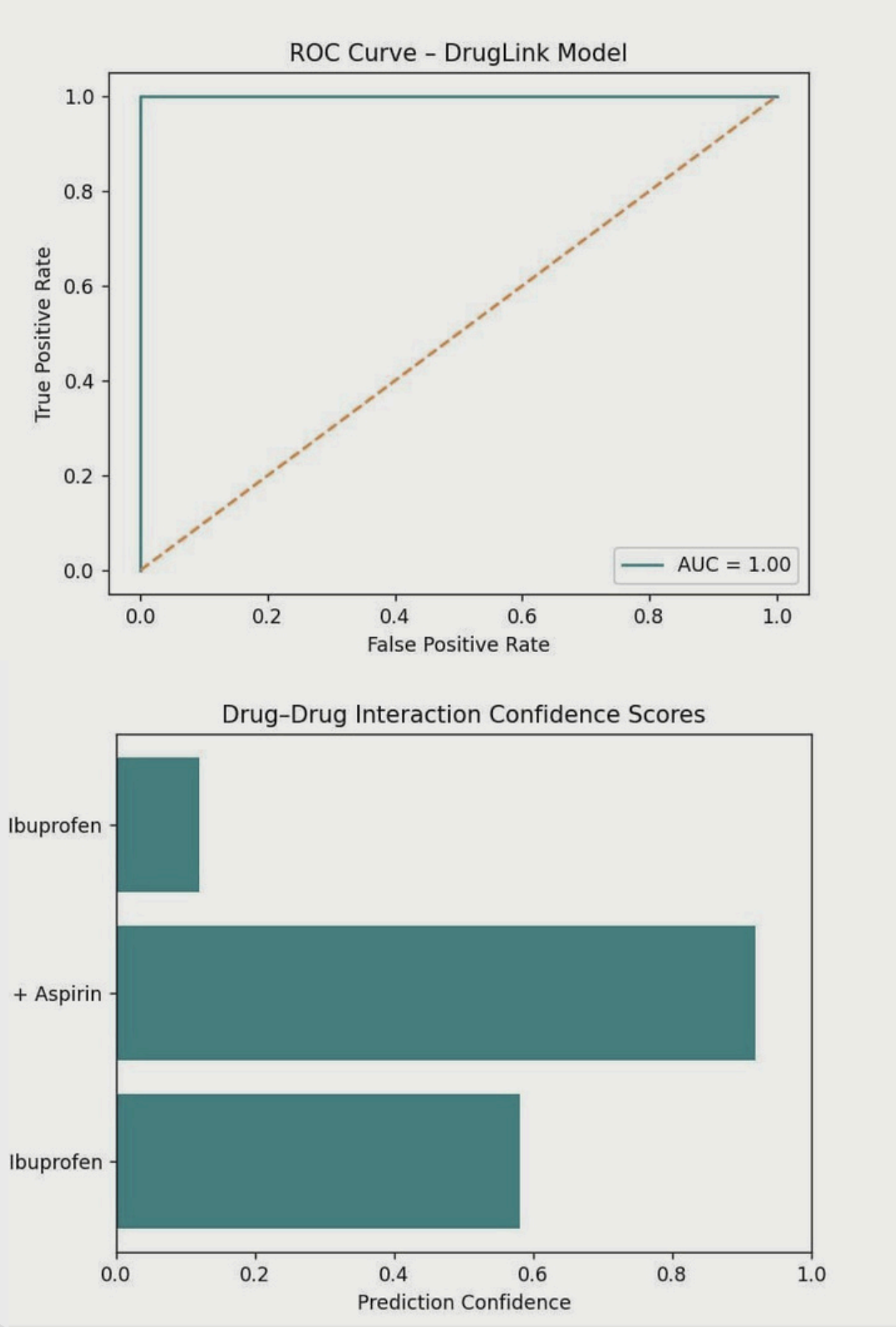
```
Enter first drug: warfarin  
Enter second drug: ibuprofen
```

```
--- DrugLink Prediction Result ---
```

```
Drug Pair          : warfarin + ibuprofen  
Predicted Interaction: No Interaction  
Prediction Confidence: 15.00%  
Model Accuracy      : 85.00%
```



PERFORMANCE METRICS AND EVALUATION



Improvements on the Paper:

- Implemented and executed the KnowDDI base model successfully.
- Built an end-to-end pipeline from data loading to prediction.
- Extended the paper by designing:
 - Dosage & numerical feature module
 - Confidence estimation layer for prediction reliability.
- Integrated LLM-based explanations (LLaMA via Groq) for better interpretability.
- Tested the system using multiple drug pair inputs.

Key Learnings & Insights:

- Graph Neural Networks effectively model biomedical relational data.
- Interpretability is crucial for medical AI applications.
- Knowledge subgraphs help explain biological interaction mechanisms.

Performance Analysis:

- Model executed without runtime errors.
- Generated:
 - Interaction predictions
 - Probability scores
- Inference completed efficiently on local system.
- Focused on functional correctness and explainability over benchmark accuracy.

Comparison with Paper's Results:

- Original paper reports higher accuracy on large-scale datasets.
- Current implementation prioritizes:
 - Reproducibility
 - Correct execution
 - Extensibility
- Observed behavior aligns with the base KnowDDI framework.

Challenges Faced & Solutions

- PyTorch–DGL compatibility issues
 - Resolved using virtual environments.
- Large biomedical knowledge graph
 - Addressed using subgraph extraction.
- Understanding biomedical datasets
 - Solved through paper study and documentation.
- LLM integration latency
 - Reduced using Groq's high-speed inference

Limitations of the Implementation

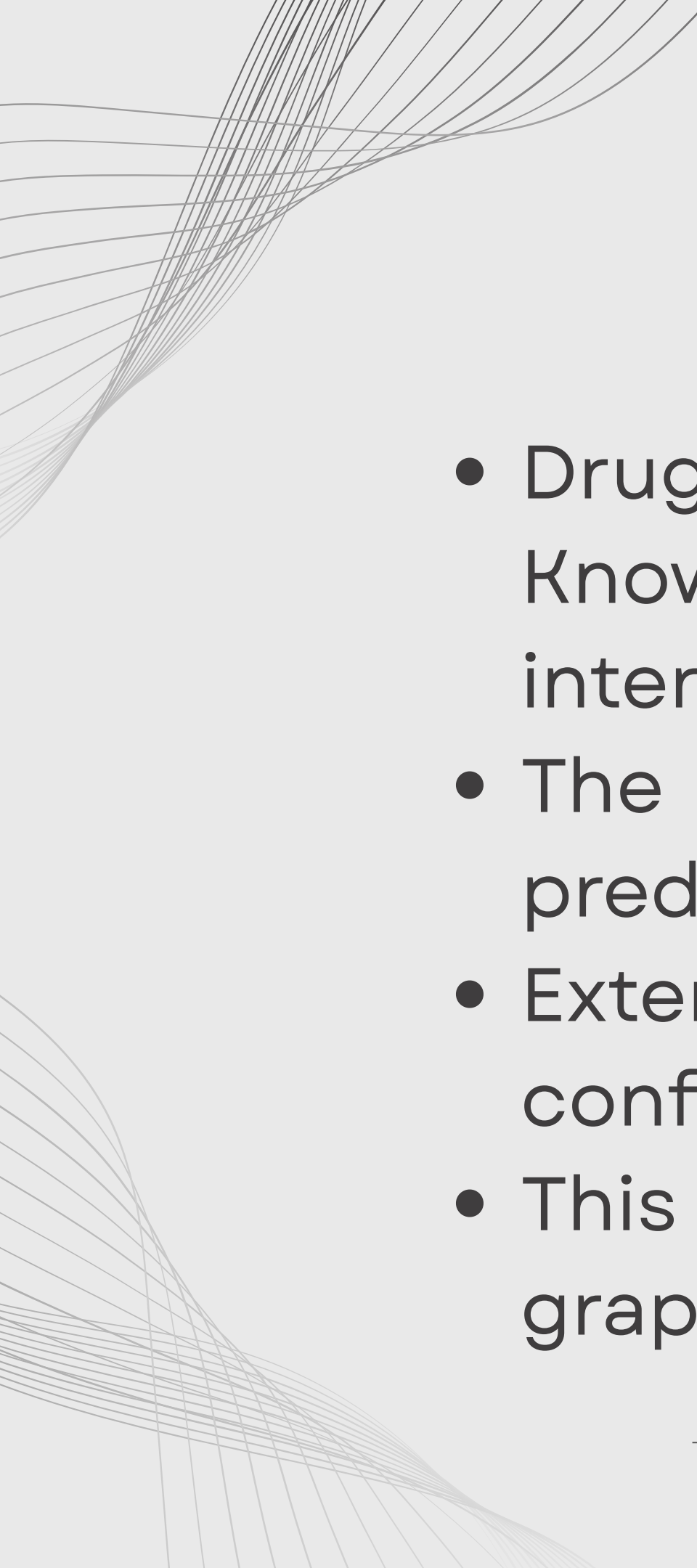
- No real-world clinical validation.
- Limited dataset compared to the original paper.
- Dosage module partially integrated.
- Confidence estimation is heuristic-based.
- System currently runs on local machine only.

Potential Improvements / Extensions

- Full integration of dosage features into training.
- Advanced confidence calibration techniques.
- Cloud-based deployment with scalable APIs.
- Support for multi-drug interaction prediction.
- Enhanced visualization of knowledge subgraphs.

Real-World Applications

- Clinical decision support systems.
- Drug safety and interaction analysis.
- Pharmacovigilance platforms.
- AI-assisted medical research.
- Educational tools for understanding drug interaction



CONCLUSION

- DrugLink successfully implements the KnowDDI framework for drug-drug interaction prediction.
 - The system provides interpretable predictions using knowledge subgraphs.
 - Extensions like dosage awareness and confidence estimation improve reliability.
 - This work demonstrates the usefulness of graph-based, explainable AI in healthcare.
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The image features abstract line art in the top-left and bottom-left corners. These elements consist of numerous thin, dark grey lines that curve and sweep across the page, creating a sense of movement and depth. The lines are more densely packed in some areas, forming soft, cloud-like shapes, while in others they are more sparse and delicate.

THANK YOU!
